

Hamiltonian Cycle Problem with Fixed Starting Vertex

Introduction

The Hamiltonian Cycle Problem is a classical graph problem in which a cycle must visit every vertex exactly once and return to the starting vertex. It has applications in route planning, scheduling, network design, and optimization problems.

Problem Description

Given a graph and a specified starting vertex, determine whether a Hamiltonian cycle exists. If a cycle exists, output the sequence of vertices forming the cycle; otherwise, report that no solution exists.

Inputs

- Graph $G(V,E)$
- Starting vertex

Output

- Hamiltonian cycle
- No solution if a cycle does not exist

Solution Approach

A backtracking algorithm is used to construct the cycle incrementally. Beginning from the specified starting vertex, the algorithm selects adjacent unvisited vertices and extends the path. If a dead end is reached, the algorithm backtracks and tries alternative vertices.

Analysis

The algorithm explores all feasible paths while using pruning to eliminate invalid paths. Vertices that have already been visited or are not adjacent to the current vertex are excluded from consideration.

Correctness

Every possible cycle starting from the specified vertex is examined. Since invalid paths are discarded and all valid paths are explored, the algorithm guarantees a correct result.

Complexity

- Time Complexity: $O(n!)$
- Space Complexity: $O(n)$

Conclusion

The Hamiltonian Cycle Problem demonstrates the effectiveness of backtracking in graph traversal problems. Although computationally expensive, the method guarantees correct identification of Hamiltonian cycles when they exist.